

Oracle 19c + APEX + ORDS + Tomcat

Installation Standard Operating Procedure

Oracle Linux 8 • On-Premises Standalone Server

Canonical Directory Structure

Use this layout consistently throughout the entire installation. Do not deviate.

/opt/app/oracle	← ORACLE_BASE
/opt/app/oracle/product/19.0.0/dbhome_1	← ORACLE_HOME (DB software)
/opt/app/oracle/oradata	← Database data files
/opt/app/oracle/apex	← APEX software
/opt/app/oracle/ords	← ORDS software
/opt/app/oracle/ords_config	← ORDS runtime configuration
/opt/app/oracle/ords_config/logs	← ORDS log files
/opt/app/oraInventory	← Oracle inventory
/opt/tomcat/apache-tomcat-9.0.x	← Tomcat home

oracle User Environment (.bash_profile)

Add the following block to /home/oracle/.bash_profile. Apply with: source ~/.bash_profile

```
# Oracle 19c environment
export ORACLE_BASE=/opt/app/oracle
export ORACLE_HOME=$ORACLE_BASE/product/19.0.0/dbhome_1
export ORACLE_SID=orcl
export PATH=$PATH:$ORACLE_HOME/bin
export CV_ASSUME_DISTID=OEL7
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:/lib:/usr/lib
export CLASSPATH=$ORACLE_HOME/jlib:$ORACLE_HOME/rdbms/jlib

# ORDS environment
export ORDS_HOME=/opt/app/oracle/ords
export ORDS_CONFIG=/opt/app/oracle/ords_config
export PATH=$ORDS_HOME/bin:$PATH

# Java 17
export JAVA_HOME=/usr/lib/jvm/java-17-openjdk
export PATH=$JAVA_HOME/bin:$PATH
```

Phase 1 — OS Prerequisites

1.1 Install Required Packages

```
sudo dnf install -y oracle-database-preinstall-19c
sudo dnf install -y unzip zip wget net-tools
sudo dnf install -y java-17-openjdk java-17-openjdk-devel
```

The preinstall package automatically creates:

- OS user: oracle
- OS groups: oinstall, dba, oper, backupdba, dgdba, kmdba, racdba
- Required kernel parameters (sysctl) and resource limits

1.2 Disable SELinux and Firewall (for installation)

```
sudo setenforce 0
sudo systemctl stop firewalld
sudo systemctl disable firewalld
```

1.3 Create Installation Directories

```
sudo mkdir -p /opt/app/oracle/product/19.0.0/dbhome_1
sudo mkdir -p /opt/app/oracle/oradata
sudo mkdir -p /opt/app/oraInventory
sudo chown -R oracle:oinstall /opt/app
sudo chmod -R 775 /opt/app
```

1.4 Set Java 17 as Default

```
sudo alternatives --config java      # select java-17-openjdk
java -version                       # confirm: openjdk version 17.x.x
```

Phase 2 — Oracle Database 19c Software Installation

2.1 Download & Extract Binaries

Download LINUX.X64_193000_db_home.zip from Oracle eDelivery / MOS, then:

```
sudo su - oracle
unzip LINUX.X64_193000_db_home.zip -d /opt/app/oracle/product/19.0.0/dbhome_1
source ~/.bash_profile
```

2.2 Run the Installer

Option A — GUI (recommended when a desktop/X11 session is available):

```
cd $ORACLE_HOME
./runInstaller
```

Follow the prompts. When asked for inventory, use /opt/app/oraInventory. Install software only (no DB creation yet).

Option B — Silent (server with no GUI):

```
$ORACLE_HOME/runInstaller -silent \
  -responseFile $ORACLE_HOME/install/response/db_install.rsp \
  oracle.install.option=INSTALL_DB_SWONLY \
  ORACLE_HOME=$ORACLE_HOME \
  ORACLE_BASE=$ORACLE_BASE \
  oracle.install.db.InstalledEdition=EE \
  oracle.install.db.OSDBA_GROUP=dba \
  oracle.install.db.OSOPER_GROUP=dba \
  oracle.install.db.OSBACKUPDBA_GROUP=dba \
  oracle.install.db.OSDGDBA_GROUP=dba \
  oracle.install.db.OSKMDBA_GROUP=dba \
  oracle.install.db.OSRACDBA_GROUP=dba \
  DECLINE_SECURITY_UPDATES=true
```

⚠ Note: If the installer fails an OS version check, ensure CV_ASSUME_DISTID=OEL7 is exported before running runInstaller.

2.3 Run Root Scripts (when prompted)

```
sudo /opt/app/oraInventory/orainstRoot.sh
sudo $ORACLE_HOME/root.sh
```

Phase 3 — Create the Database

3.1 Create Database Using DBCA (Silent)

```
dbca -silent -createDatabase \
  -templateName General_Purpose.dbc \
  -gdbname orcl \
  -sid orcl \
  -responseFile NO_VALUE \
  -characterSet AL32UTF8 \
  -sysPassword SysPassword123 \
  -systemPassword SysPassword123 \
  -createAsContainerDatabase false \
  -datafileDestination "/opt/app/oracle/oradata"
```

3.2 Verify Database & Listener

```
lsnrctl status
sqlplus / as sysdba
  startup;                                -- if instance is idle
  alter system register;                  -- register with listener
  select name from v$database;
  select instance_name from v$instance;
  show parameter service_name;
```

3.3 Verify tnsnames.ora

File location: \$ORACLE_HOME/network/admin/tnsnames.ora

```
ORCL =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = localhost) (PORT = 1521))
    (CONNECT_DATA =
      (SERVICE_NAME = orcl)
    )
  )
```

⚠ Note: If the service does not register with the listener, run: `ALTER SYSTEM SET local_listener='(ADDRESS=(PROTOCOL=TCP)(HOST=localhost)(PORT=1521))' SCOPE=BOTH;` then `ALTER SYSTEM REGISTER;`

Phase 4 — Install Oracle APEX

4.1 Download & Extract

Download apex-latest.zip from <https://apex.oracle.com/en/download/> (free, no account required).

```
cd /opt/app/oracle
unzip apex-latest.zip
# Result: /opt/app/oracle/apex
sudo chown -R oracle:oinstall /opt/app/oracle/apex
```

4.2 Install APEX into the Database

```
sqlplus / as sysdba
startup; -- ensure DB is open

cd /opt/app/oracle/apex
sqlplus / as sysdba
@apexins.sql SYSAUX SYSAUX TEMP /i/
```

Installation takes 10–20 minutes. Parameters: tablespace = SYSAUX, temp = TEMP, images virtual path = /i/

4.3 Set APEX Admin Password

```
@apxchpwd.sql
```

Sets the password for the APEX ADMIN workspace account.

4.4 Configure APEX REST (required for ORDS)

```
@apex_rest_config.sql
```

Sets passwords for APEX_PUBLIC_USER and APEX_LISTENER. Record these passwords — they are needed during ORDS installation.

4.5 Unlock APEX_PUBLIC_USER

```
sqlplus / as sysdba
ALTER USER APEX_PUBLIC_USER ACCOUNT UNLOCK;
ALTER USER APEX_PUBLIC_USER IDENTIFIED BY "YourApexPublicPassword";
```

Phase 5 — Install Oracle REST Data Services (ORDS)

5.1 Download & Extract

Download ords-latest.zip from <https://www.oracle.com/tools/downloads/rest-data-services-downloads.html>

```
sudo mkdir -p /opt/app/oracle/ords
sudo chown -R oracle:oinstall /opt/app/oracle/ords
cd /opt/app/oracle/ords
unzip /path/to/ords-latest.zip
# Verify: ls bin/ords (must exist)
```

5.2 Create Config Directories

```
mkdir -p /opt/app/oracle/ords_config
mkdir -p /opt/app/oracle/ords_config/logs
```

5.3 Silent Installation Script

Create the script file:

```
vim /opt/app/oracle/install_ords_silent.sh
```

Paste the following content (update passwords as needed):

```
#!/bin/bash
export ORDS_HOME=/opt/app/oracle/ords
export ORDS_CONFIG=/opt/app/oracle/ords_config
export ORDS_LOGS=${ORDS_CONFIG}/logs
export DB_HOSTNAME=localhost
export DB_PORT=1521
export DB_SERVICE=orcl
export SYSDBA_USER=SYS
export SYSDBA_PASSWORD=YourSysPassword
export ORDS_PASSWORD=YourOrdsPassword

${ORDS_HOME}/bin/ords --config ${ORDS_CONFIG} install \
  --log-folder ${ORDS_LOGS} \
  --admin-user ${SYSDBA_USER} \
  --db-hostname ${DB_HOSTNAME} \
  --db-port ${DB_PORT} \
  --db-servicename ${DB_SERVICE} \
  --feature-db-api true \
  --feature-rest-enabled-sql true \
  --feature-sdw true \
  --gateway-mode proxied \
  --gateway-user APEX_PUBLIC_USER \
```

```
--proxy-user \
--password-stdin <<EOF
${SYSDBA_PASSWORD}
${ORDS_PASSWORD}
EOF
```

```
chmod +x /opt/app/oracle/install_ords_silent.sh
/opt/app/oracle/install_ords_silent.sh
```

Expected output:

- Configuring database...
- Creating ORDS metadata...
- Creating ORDS_PUBLIC_USER...
- Installation completed.

5.4 Unlock ORDS_PUBLIC_USER (if needed)

```
sqlplus / as sysdba
ALTER USER ORDS_PUBLIC_USER ACCOUNT UNLOCK;
ALTER USER ORDS_PUBLIC_USER IDENTIFIED BY "StrongPasswordHere";
```

5.5 Generate ORDS WAR File for Tomcat

```
cd /opt/app/oracle/ords/bin
./ords --config /opt/app/oracle/ords_config war ords.war
```

Expected output: WAR generation complete. The file is created at /opt/app/oracle/ords/bin/ords.war

Phase 6 — Install Apache Tomcat

6.1 Download & Extract

```
sudo mkdir -p /opt/tomcat
sudo chown oracle:oinstall /opt/tomcat
cd /opt/tomcat
```

Download apache-tomcat-9.0.x.tar.gz from <https://tomcat.apache.org> then:

```
tar -xvf apache-tomcat-9.0.x.tar.gz
# Result: /opt/tomcat/apache-tomcat-9.0.x
chmod +x /opt/tomcat/apache-tomcat-9.0.x/bin/*.sh
```

6.2 Change Tomcat Port (avoid conflict with ORDS standalone)

Edit /opt/tomcat/apache-tomcat-9.0.x/conf/server.xml. Find:

```
<Connector port="8080" protocol="HTTP/1.1"
```

Change to port 8085 (or another free port):

```
<Connector port="8085" protocol="HTTP/1.1"
```

6.3 Create setenv.sh for ORDS Config Path

```
vim /opt/tomcat/apache-tomcat-9.0.x/bin/setenv.sh
```

```
#!/bin/bash
export ORDS_CONFIG=/opt/app/oracle/ords_config
export JAVA_OPTS="$JAVA_OPTS -Dconfig.dir=$ORDS_CONFIG"
export CATALINA_OPTS="$CATALINA_OPTS -Dconfig.dir=$ORDS_CONFIG"
```

```
chmod +x /opt/tomcat/apache-tomcat-9.0.x/bin/setenv.sh
```

6.4 Create Systemd Service

```
sudo nano /etc/systemd/system/tomcat.service
```

```
[Unit]
Description=Apache Tomcat Application Server
After=network.target

[Service]
Type=forking
User=oracle
Group=oinstall
WorkingDirectory=/opt/tomcat/apache-tomcat-9.0.x
Environment="JAVA_HOME=/usr/lib/jvm/java-17-openjdk"
Environment="CATALINA_HOME=/opt/tomcat/apache-tomcat-9.0.x"
Environment="CATALINA_BASE=/opt/tomcat/apache-tomcat-9.0.x"
ExecStart=/opt/tomcat/apache-tomcat-9.0.x/bin/startup.sh
ExecStop=/opt/tomcat/apache-tomcat-9.0.x/bin/shutdown.sh
TimeoutStopSec=30
Restart=on-failure

[Install]
WantedBy=multi-user.target
```

```
sudo systemctl daemon-reload
sudo systemctl enable tomcat
sudo systemctl start tomcat
sudo systemctl status tomcat
```

6.5 Deploy ORDS WAR to Tomcat

```
cp /opt/app/oracle/ords/bin/ords.war /opt/tomcat/apache-tomcat-9.0.x/webapps/
sudo systemctl restart tomcat
```

Tomcat auto-unpacks the WAR. Verify at: <http://<server-ip>:8085/ords/>

Phase 7 — Verification & Troubleshooting

7.1 End-to-End Verification Checklist

- Database listener running: lsnrctl status
- Database open: sqlplus / as sysdba → select status from v\$instance;
- Service registered: lsnrctl status shows 'Service orcl ... status READY'
- Tomcat running: systemctl status tomcat
- APEX accessible: http://<server-ip>:8085/ords/apex

7.2 Common Issues

Listener: service not registered

```
sqlplus / as sysdba
ALTER SYSTEM SET local_listener=
  '(ADDRESS=(PROTOCOL=TCP) (HOST=localhost) (PORT=1521))' SCOPE=BOTH;
ALTER SYSTEM REGISTER;
lsnrctl status
```

SELinux blocking Tomcat

```
getenforce
sudo setenforce 0
sudo systemctl restart tomcat
```

Port conflict on 8080

```
sudo lsof -i :8080
```

If ORDS standalone is on 8080, change Tomcat to 8085 in server.xml (see Step 6.2).

Wrong Java version

```
java -version          # must show 17.x.x
sudo alternatives --config java  # select java-17-openjdk
sudo systemctl daemon-reload
sudo systemctl restart tomcat
```

ORDS_PUBLIC_USER locked

```
sqlplus / as sysdba
ALTER USER ORDS_PUBLIC_USER ACCOUNT UNLOCK;
ALTER USER ORDS_PUBLIC_USER IDENTIFIED BY "NewPassword";
```

7.3 Useful Diagnostic Commands

```
# Check all Oracle-related processes
ps -ef | grep -i oracle

# Verify ORDS config
ords --config /opt/app/oracle/ords_config config list

# Tomcat logs
```



```
tail -100 /opt/tomcat/apache-tomcat-9.0.x/logs/catalina.out
```

```
# ORDS logs
```

```
tail -100 /opt/app/oracle/ords_config/logs/ords.log
```